A
B. Tech (CSE)

PROJECT REPORT ON

"Canteen Management System"

Submitted by

Jay Ingle – 202356108708

Uday Salve – 202256108037

Vaibhav Chaudhary – 202256108029

Parth Kulkarni - 202356108703.

Under Guidance of

Ms. Jadhav A.K

School of Engineering and Technology(SOET)
MGM University, Chh.Sambhajinagar
Academic Year 2025-2026

School of Engineering and Technology

MGM University, Chh. Sambhajinagar

CERTIFICATE

This is to certify that Jay Ingle, Uday Salve, Vaibhav Chaudhary, Parth Kulkarni of B.Tech (IoT, Cyber Security Including Blockchain Technology), PRN No. 202356108708, 202256108037, 202256108029, 202356108703 has successfully completed Project work on **"Canteen Management System"** under the guidance of **Ms. Jadhav A.K** and submitted the same during the academic year 2025- 2026 towards the partial fulfillment of degree of B.Tech (IoT, Cyber Security Including Blockchain Technology) from School of Engineering and Technology, MGM University, Chh.Sambhajinagr.

Ms. Jadhav A.K **Ms.Rathi S.R**

Project Guide **HOD(CSE)**

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Jay Ingle - 202356108708

Uday Salve - 202256108037

Vaibhav Chaudhary - 202256108029

Parth Kulkarni - 202356108703.

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Abstract

The Canteen Management System (CMS) is a web-based application designed to automate and streamline the operations of a modern canteen. Traditional canteen processes, such as manual order taking, billing, and inventory tracking, are often slow, error-prone, and inefficient. Canteen Management System addresses these challenges by providing a fully digital platform that includes separate interfaces for customers and administrators. A complete full-stack functional Canteen Management System with a modern UI, real-time functionality, and many other features for both customer and administrator sides.Customers can browse a dynamic menu, place orders, and make payments through wallet or online options, while administrators can manage the menu, monitor user activity, and generate detailed financial reports. The system leverages HTML, CSS, JavaScript, and localStorage to maintain data persistence, ensuring reliability and portability without the need for a backend server in the prototype stage. Key Features: Menu display, shopping cart, user authentication, customer and admin roles, ordering, order management, statistics dashboard. Architectural Scalability and Robustness .The backend logic uses the fast Node.js runtime and the expressive Express.js framework for full power on routing and API; and for the purpose of prototyping, data persistence including user profile, menu, and order history is managed using browser localstorage and a JSON data structure which means that there is no need for external server facilities, making it both portable and quick to deploy; the whole system is inherently scalable; it is designed from the outset to easily incorporate enterprise-grade databases (such as PostgreSQL) and microservices architecture, future-proofing the CMS for increasing user engagement, scaling to an increasing feature set, and automating the billing and inventory tracking as well.

Abbreviations

CMS: Canteen Management System.

HTML: Hyper Text Markup Language.

CSS: Cascading Style Sheets.

CSV : Comma Separated File.

DOM : Document Object Model .

JS/ES6+: JavaScript/ECMA Script 2015+.

API: Application Programming Interface, REST- full API Design.

KDS : Kitchen Display System .

CRUD: Create, Read, Update, Delete .

CORS: Cross-Origin Resource Sharing.

ADS: Admin Display System.

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Chapter 1: Introduction

1.1 Introduction

The core requirement for digitalization in canteen operations is to enhance efficiency and the customer experience. Long queues during peak time, errors in manual billing, sluggish tracking of inventory, and delayed financial reporting are common problems in traditional canteen systems. In turn, CMS will be designed to solve these challenges through complete digital automation. The system will be a full-stack web application with a modern, user-friendly interface. The modern canteen environment demands not just speed but also precision and efficiency to meet the expectations of today's users. The core requirement for the digitalization of canteen operations is a dual focus: significantly enhancing operational efficiency and delivering a superior customer experience. This system, distinguished by its modern, user-friendly interface, will not merely digitize existing processes but fundamentally transform the canteen's operation, ensuring rapid transactions, real-time inventory oversight, and instant data analysis, thereby guaranteeing a more streamlined, reliable, and satisfactory service for every user.

1.2 Problem Statement

Traditional manual ordering systems are slow and error-prone, with no real-time data. The CMS resolves these issues by offering a digital avenue for order processing and management.The current manual system poses severe operational difficulties, which seriously compromise both efficiency and financial accuracy. The process is tediously long, leading to long queues for customers where orders have to be taken manually and bills provided. Secondly, the system is very prone to errors, often leading to wrong order fillings, incorrect customer billing, and miscalculations in key daily sales reports. From an operational perspective, the managers are critically deprived of control, in as much as there is no real capability for real-time monitoring of key data points such as menu stock, customer spending, and status of current orders. Traditional, manual methods currently employed in canteen operations result in significant operational bottlenecks and a compromised user experience.

The absence of an integrated, automated system directly leads to predictable and detrimental inefficiencies that negatively impact financial performance, administrative control, and customer satisfaction.

1.3 Objectives

The core objective of this project is to deliver a comprehensive, responsive, web-based ordering system with a separate interface for both the customer and administrator use cases. The Customer Self-Service module allows users to view the menu, place orders, and choose their payment method, with a strong focus on cashless transactions through the use of integrated wallet features and a simulated Razorpay payment gateway. In addition, the Admin Interface offers real-time order tracking, thereby enabling staff to manage the status of an order as New, Ready, and Complete for efficient workflow management.

1.4 Scope of the Project

The Canteen Management System is designed to facilitate the automation of all the crucial operations and management of a modern canteen. The project includes functionalities found in Customer Portal, Admin Portal-admin_portal.html, and Transaction Records page-admin_transactions.html.

This project develops a fully functional Digital Canteen Management System, focusing on the enhancement of customer experience and administrative control. The core feature of Customer Self-Service can be accessed through c3.html, thus enabling the users to perform registration, login, menu browsing, cart management, and order finalization with ease. Complementing this front-end, the system incorporates several Cashless Transaction possibilities: wallet deduction, online payment simulation, and traditional cash-at-counter. Importantly, administrative efficiency is ensured through a dedicated Admin Portal capable of performing Real-time Order Tracking and status updating. This portal also provides strong Data Management functionalities for full CRUD operations on menu items and user accounts.

Chapter 2: Literature Review

2.1 Review of Existing System / Related Work

2.1.1 Review of the Existing System (Manual Canteen Operations)

The current operational framework for canteen management is a Traditional Manual System, characterized by its heavy reliance on labor-intensive, paper-based processes which form the critical justification for the CMS implementation. This methodology is inherently inefficient: order taking is slow and manual, directly causing long customer queues during peak times. Financial operations lack control and transparency, as handwritten, cash-based billing significantly increases the risk of human error, leading to frequent discrepancies and financial losses, while also raising hygiene concerns and complicating daily reconciliation. From an administrative standpoint, the system is wholly inadequate, as vital operational data (sales, inventory, transactions) is scattered across physical ledgers, making report generation tedious, time-consuming, and vulnerable to data loss. Managers are consequently deprived of real-time visibility, hindering their ability to make informed decisions regarding stock and menu adjustments. In summary, the existing manual system is defined by its inefficiency, inaccuracy, security risks, and lack of scalability, underscoring the necessity of the proposed digital Canteen Management System (CMS) to overcome these constraints through automation, data centralization, and the provision of faster, more accurate, and completely transparent service.

2.1.2 Related Work Review - Digital Ordering Solutions

The review of existing systems confirms that the digital transition is an industry standard, where most modern quick-service restaurants and institutional canteens have successfully adopted sophisticated digital ordering solutions. These systems universally feature integrated web and mobile ordering applications alongside robust digital payment solutions, proving their capability to deliver seamless operations and a superior customer experience. The proposed Canteen Management System (CMS) directly applies these successful industry concepts by incorporating a dedicated customer self-ordering web

portal and facilitating modern payment methods through e-wallet integration and online payment simulations. This strategic alignment ensures that the CMS is engineered to provide faster transactions, significantly improved order accuracy, and a seamless digital experience that meets contemporary industry performance and user expectations.

2.2 Knowledge Gap

The transition from the existing manual canteen operation to the proposed Canteen Management System (CMS) is driven by a critical knowledge gap rooted in fundamental deficiencies in the manual process across four key areas: data handling, speed, scalability, and customer engagement. Manual systems inherently lack data integrity and real-time visibility, making order processing inefficient and financial reporting slow and unreliable. Specifically, the processes for financial reconciliation are tediously time-consuming, and the reliance on cash-only transactions severely limits customer convenience and operational transparency. The proposed CMS directly bridges this gap through complete digital automation, ensuring robust data consistency, enabling instant order processing and real-time financial monitoring, and fundamentally enhancing both operational efficiency and scalability through the integration of digital e-wallet payments and automated reporting features, resulting in a significantly modern user experience.

2.3 Summary of Findings

The Digital Canteen Management System effectively bridges the operational and The Digital Canteen Management System (CMS) conclusively bridges the operational and technological gaps identified in the manual process. Through complete automation, the system effectively reduces order waiting time and eliminates common billing errors, ensuring faster, more accurate service delivery. Operationally, the CMS promotes a modern, efficient environment by supporting cashless, hygienic transactions and centralizing data management. These enhancements provide significant benefits to the user base, including personalized features like order history and future potential for reward points, all while guaranteeing a significantly faster service delivery experience compared to the traditional system.

Chapter 3: System Analysis

3.1 Requirement Analysis

3.1.1 Functional Requirements:

The project delivers an end-to-end digital restaurant management system, making the operational workflow smooth, from placing orders to advanced business intelligence. It includes secure authentication of customers and admins via their respective login/signup portals, efficient Menu Management for adding or removing items in the cart, and robust functionality related to Order Fulfillment supported by real-time updates and tracking via the Kitchen Display System. Financial efficiency in the wallet for cashless transactions will be maintained with features like recharge and deduction, and the system will simulate external transactions through Razorpay. Vital data governance is ensured through CSV export of data for detailed transactions and sales reports. Crucially, the system will come with an AI Forecasting module, which can handle the forecast of top-selling items, thereby analyzing the revenue trend for delivering actionable, data-driven insights in strategic planning for business growth.

3.1.2 Non-Functional Requirements:

The application architecture was developed with a strong focus on core quality attributes, prioritizing robust performance and security. Performance is ensured through the implementation of fast loading techniques, asynchronous operations, and efficient Document Object Model (DOM) updates. For security, the system enforces rigorous session validation, comprehensive input sanitization, and secure management of local or session storage to protect user data and maintain integrity. Furthermore, the overall design emphasizes usability, providing a clean and intuitive user interface across both the Customer and Admin portals. This commitment extends to full responsiveness, guaranteeing a seamless and optimal experience across all device types, including desktop, tablet, and mobile platforms.

3.2 Feasibility Study

3.2.1 Technical Feasibility:

The project demonstrates strong technical and economic feasibility. Technically, the system is built upon a mature, open-source web stack utilizing HTML5, CSS3, JavaScript, Node.js, and Express.js, ensuring long-term stability and broad development support. Data persistence is managed efficiently via local storage, which eliminates the need for complex server setups and simplifies the maintenance, deployment, and scaling processes for individuals with basic web development skills. Economically, the solution is highly attractive, requiring very low development and maintenance costs with no anticipated recurring expenses. This low overhead, combined with the high potential Return on Investment (ROI) driven by improved order processing speed and enhanced billing accuracy, validates the project's financial viability.

3.2.3 Operational Feasibility:

Our comprehensive digital solution is built around a user-friendly, mobile-responsive design to guarantee high customer acceptance and a superior ordering experience. Operationally, the system ensures a smooth administrative workflow by utilizing a digital Kitchen Display System (KDS) and providing real-time status updates. For financial management, it seamlessly supports multiple payment options while meticulously recording all transactions. Furthermore, the integrated AI Forecasting feature provides critical assistance in inventory planning and proactively reduces wastage, driving overall business efficiency.

3.3 System Specifications

3.3.1 Hardware Requirements:

• Standard desktop/laptop or smartphone with at least 4GB RAM.

• Any modern web browser: Chrome, Firefox, Edge.

3.3.2 Software Requirements:

• Front-end: HTML5, CSS3, JavaScript (ES6+).

• Back-end: Node.js with Express.js framework.

• Database Simulation: Browser localStorage API.

• External Tools: Font Awesome, Razorpay SDK, VS Code, Git.

Chapter 4: System Design

4.1 System Architecture

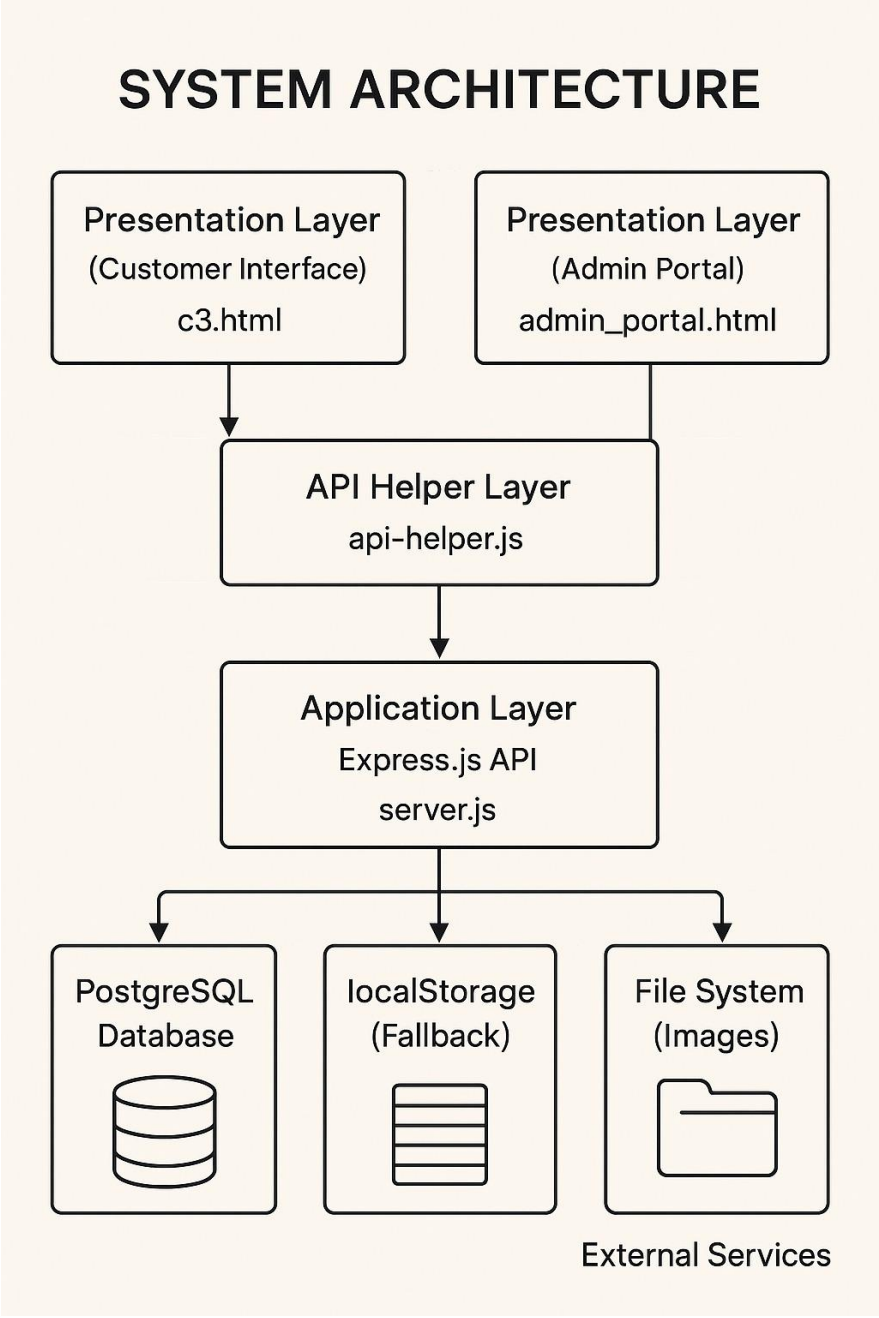


Figure 4.1.1:-System Architecture

4.2 Data Flow Diagram/ UML Diagram

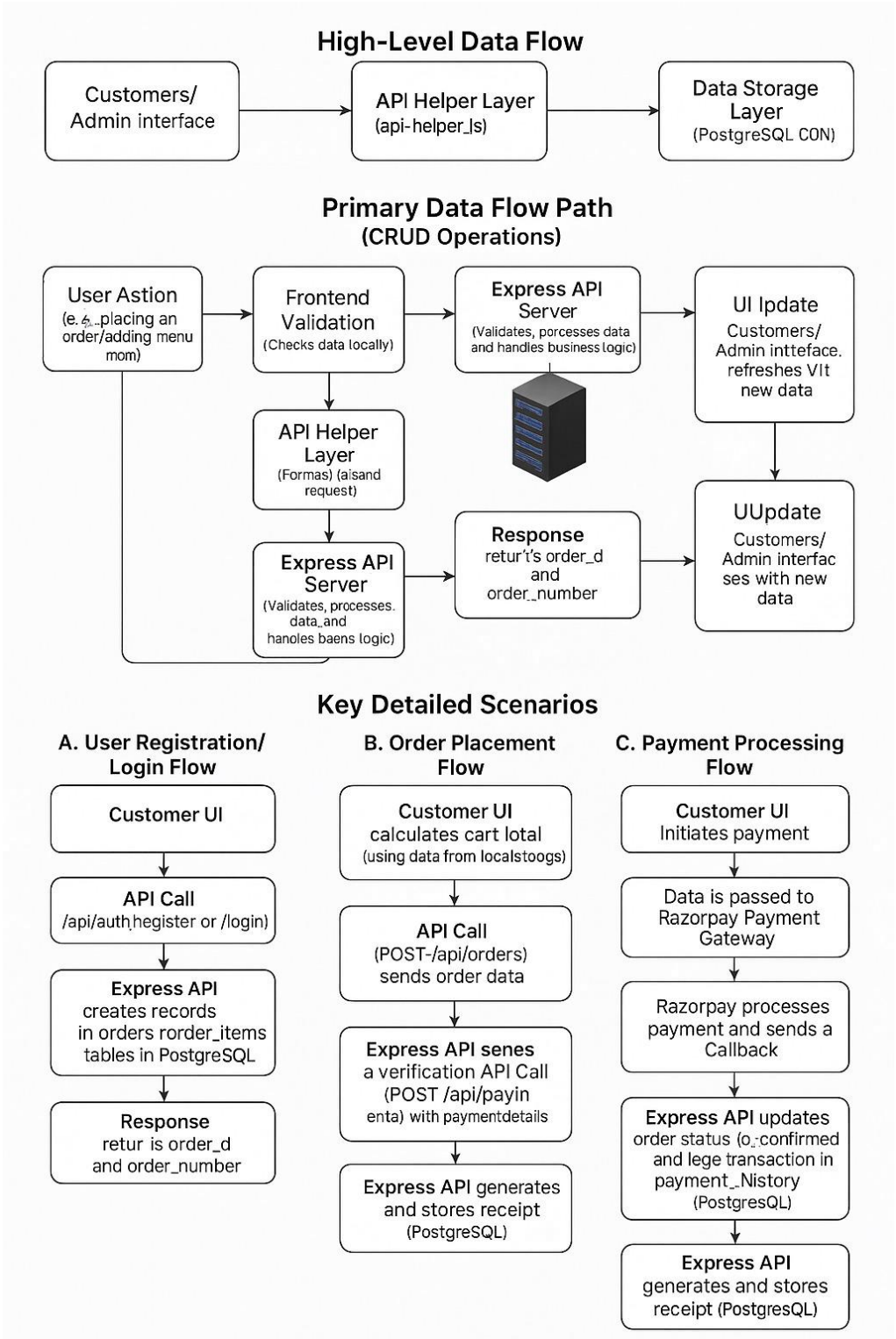


Figure 4.2.1:-Data Flow Diagram/ UML Diagram

4.3 ER Diagram / Database Design

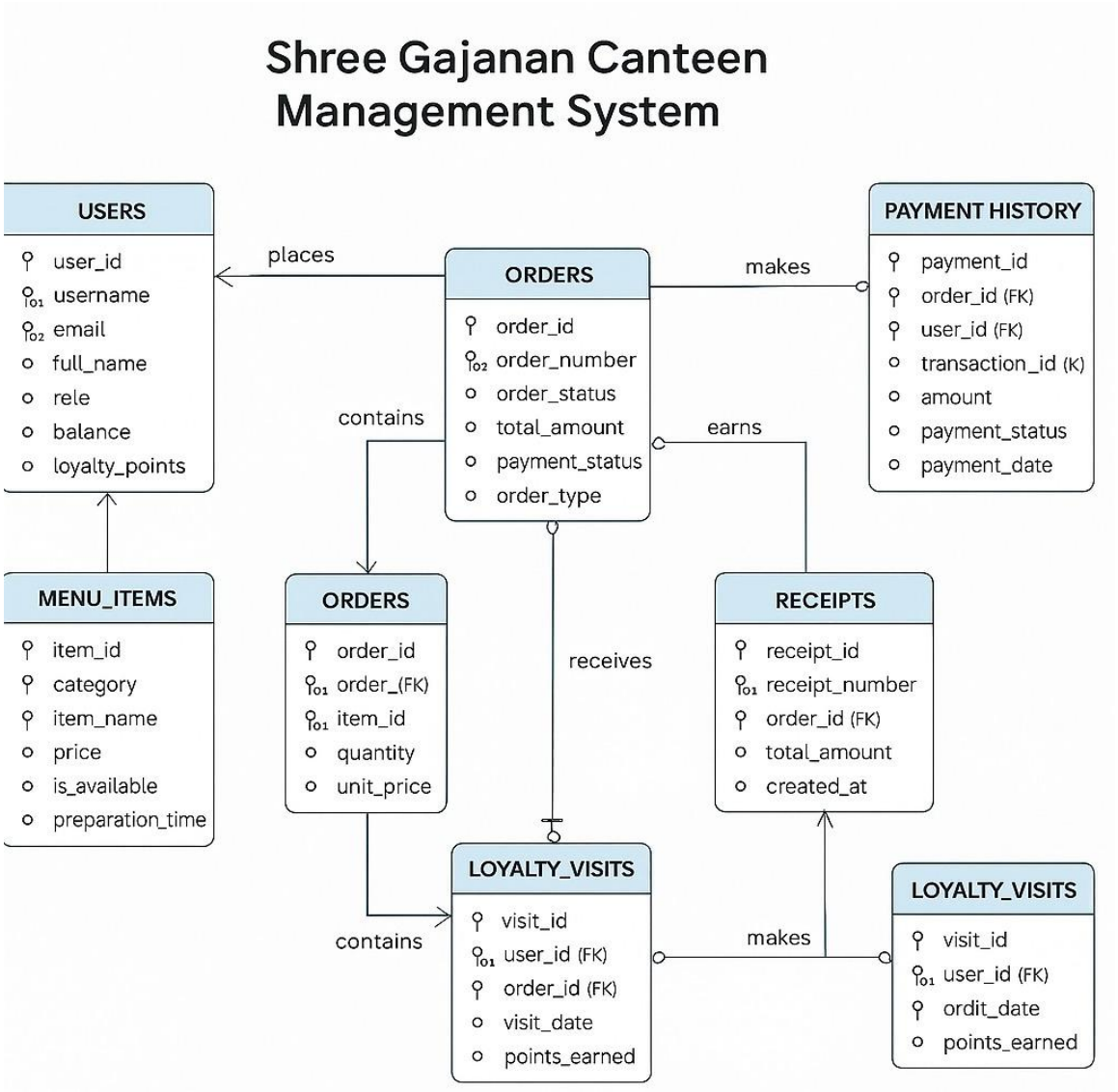


Figure 4.3.1:- ER Diagram / Database Design

Chapter 5: Implementation

5.1 Module-wise Description

The Canteen Management System (CMS) offers a set of applications or modules that automate the ordering, payment, monitoring, and reporting processes for efficient operations of the canteen. The three primary modules of the Canteen Management System include:

Customer Module.

This module begins with robust user authentication, enabling customers to utilize secure login and registration processes to access their personalized panel. Once authenticated, the module presents a dynamic menu where food items are intuitively organized into categories, making browsing and selection easy and quick, thus ensuring a minimum learning curve. A core feature is the Cart Management system, which allows users to easily add, modify, or remove items before finalizing their purchase. At the point of checkout, the module offers flexible payment method selection, supporting traditional cash-at-counter payments, deductions from a secure in-system wallet payment balance, and simulated online payment via the Razorpay integration. Furthermore, to enhance user engagement and transparency, customers can access a detailed history to view their past orders, aiding in tracking spending and identifying favorites. Crucially, the entire application interface is engineered for a superior user experience, built to be simple, responsive, and mobile-optimized, ensuring optimal functionality and visual clarity across all devices, from desktop to smartphone, thereby guaranteeing a fast and error-free transaction flow.

Admin Core Module.

The Admin Core Module serves as the central operational hub and control center for the entire Canteen Management System (CMS), designed to empower management with complete oversight and real-time control over all system functions.

This module is anchored by a comprehensive Dashboard that provides key operational intelligence at a glance, displaying real-time metrics such as daily sales figures, trending popular item selections, and the count of active users. Beyond mere monitoring, the module is equipped with robust Data Management functionalities supporting full CRUD (Create, Read, Update, Delete) operations. This allows administrators to directly manage core data entities, including updating the menu, handling user account details, controlling digital wallets (for crediting/debiting), viewing detailed transaction logs, and managing any promotional elements like gift cards. Crucially, the Admin Core Module integrates a simulated AI-based Forecasting System. Leveraging historical sales data (as outlined in the Appendices), this system provides predictive business intelligence by identifying the top seller items and estimating revenue projections for the following day, thereby enabling proactive, data-driven decisions on inventory ordering and preparation, directly solving the problem of sluggish inventory tracking highlighted in the Problem Statement.

Transaction / Report Module

The Transaction / Reports Module has record of all financial transactions. A feature of the Transaction Module is the ability to export all financial data in a CSV format for auditing purposes. The Transaction / Reports Module has insights of financial trends - Overall revenue trends and payment trends. Transparency is built into the system to allow the canteen manager to effectively manage their financial obligations. Maintains a complete record of all financial transactions related to orders and payments. It also supports exporting a CSV file of the transaction log for auditing purposes.

5.2 Technologies Used

The CMS uses modern web technologies: HTML5 lays the foundation for web pages, enabling functions like structuring elements such as a form, menus, and buttons to present information to users in a clear and accessible way.CSS3 improves looks and responsive design so that the interface works across devices, utilizing things like flexbox, grid layouts, and media queries.JavaScript (ES6+) provides logic and interactivity, managing what users do like updating their cart, payments, and changing dynamic menus. Modern JavaScript

ES6 syntax, like arrow functions and modular code, enables efficiency and easier maintenance.Local Storage acts as a light client-side database, storing things like user profiles, menus, and orders in a way to give persistence without the need for a backend server, while we are in the prototype stage. Razorpay Checkout simulates a secure online payment system, giving users an experience of a digital transaction scam while illustrating how this can work once we have a payments gateway working.

5.3 Important Code Snippets and Functions

The major functionalities of the Canteen Management System deal with data storage, user.

InitDB() initializes local Storage with default menu and admin information.

it updates and stores modified data in local Storage.

```
-- ============================================
-- SAMPLE DATA (Optional - for testing)
-- ============================================

-- Insert admin user (password: admin123 - should be hashed in production)

INSERT INTO users (username, password_hash, full_name, email, phone, role, balance) VALUES
('admin', '$2b$10$rKvVLZ5Z5Z5Z5Z5Z5Z5uXxXxXxXxXxXxXxXxXxXxXxXxXxXxXxX', 'Admin User', 'admin@canteen.com', '9999999999', 'admin', 0.00);
-- Insert sample customers

INSERT INTO users (username, password_hash, full_name, email, phone, role, balance, loyalty_points) VALUES
('john_doe', '$2b$10$hashedpassword1', 'John Doe', 'john@example.com', '9876543210', 'customer', 500.00, 5),
('jane_smith', '$2b$10$hashedpassword2', 'Jane Smith', 'jane@example.com', '9876543211', 'customer', 300.00, 3),
('raj_kumar', '$2b$10$hashedpassword3', 'Raj Kumar', 'raj@example.com', '9876543212', 'customer', 750.00, 8);
-- Insert sample menu items

INSERT INTO menu_items (category, item_name, description, price, image_url, is_vegetarian) VALUES
('Tea, Coffee & Milk', 'Tea', 'Classic Indian chai', 10.00, 'chai.jpg', TRUE),
('Tea, Coffee & Milk', 'Special Tea', 'Premium masala chai', 15.00, 'special_chai.jpg', TRUE),
('Tea, Coffee & Milk', 'Coffee', 'Fresh brewed coffee', 15.00, 'coffee.jpg', TRUE),
('Tea, Coffee & Milk', 'Cold Chocolate Milk', 'Chilled chocolate milk', 25.00, 'cold_chocolate_milk.jpg', TRUE),
('Breakfast', 'Pohe', 'Traditional flattened rice', 25.00, 'pohe.jpg', TRUE),
('Breakfast', 'Vada Pav', 'Mumbai street food special', 15.00, 'vada_pav.jpg', TRUE),
('Breakfast', 'Idli Sambar', 'South Indian steamed rice cakes', 40.00, 'idli_sambar.jpg', TRUE),
('Specials & Main Course', 'Paneer Masala', 'Cottage cheese in rich gravy', 120.00, 'paneer_masala.jpg', TRUE),
('Specials & Main Course', 'Dal Fry', 'Tempered lentils', 50.00, 'dal_fry.jpg', TRUE),
('Specials & Main Course', 'Paneer Pulao', 'Fragrant rice with cottage cheese', 120.00, 'paneer_pulao.jpg', TRUE);
```

Screenshot 5.3.1 :- Login/ Signup database.

```sql
-- Shree Gajanan Canteen Database Schema
-- PostgreSQL Database
-- ============================================
    Ctrl+I for Command, Ctrl+L for Cascade
DROP TABLE IF EXISTS receipts CASCADE;
DROP TABLE IF EXISTS payment_history CASCADE;
DROP TABLE IF EXISTS order_items CASCADE;
DROP TABLE IF EXISTS orders CASCADE;
DROP TABLE IF EXISTS menu_items CASCADE;
DROP TABLE IF EXISTS users CASCADE;

-- ============================================
-- 1. USERS TABLE
-- ============================================
CREATE TABLE users (
    user_id SERIAL PRIMARY KEY,
    username VARCHAR(50) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE,
    phone VARCHAR(15),
    role VARCHAR(20) DEFAULT 'customer' CHECK (role IN ('customer', 'admin', 'staff')),
    balance DECIMAL(10, 2) DEFAULT 0.00,
    loyalty_points INTEGER DEFAULT 0,
    has_unclaimed_reward BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    last_login TIMESTAMP,
    is_active BOOLEAN DEFAULT TRUE
);

-- Create index for faster lookups
CREATE INDEX idx_users_username ON users(username);
CREATE INDEX idx_users_email ON users(email);
CREATE INDEX idx_users_role ON users(role);
```

Screenshot 5.3.2 :- User table Database.

```sql
CREATE TABLE menu_items (
    item_id SERIAL PRIMARY KEY,
    category VARCHAR(100) NOT NULL,
    item_name VARCHAR(150) NOT NULL,
    description TEXT,
    price DECIMAL(10, 2) NOT NULL,
    image_url VARCHAR(255),
    is_available BOOLEAN DEFAULT TRUE,
    is_vegetarian BOOLEAN DEFAULT TRUE,
    preparation_time INTEGER DEFAULT 10, -- in minutes
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
-- Create index for category filtering
CREATE INDEX idx_menu_category ON menu_items(category);
CREATE INDEX idx_menu_available ON menu_items(is_available);
-- ============================================
-- 3. ORDERS TABLE
-- ============================================
CREATE TABLE orders (
    order_id SERIAL PRIMARY KEY,
    user_id INTEGER REFERENCES users(user_id) ON DELETE SET NULL,
    order_number VARCHAR(20) UNIQUE NOT NULL,
    table_number INTEGER,
    order_status VARCHAR(20) DEFAULT 'pending' CHECK (order_status IN ('pending', 'confirmed', 'preparing', 'ready', 'completed', 'cancell
    subtotal DECIMAL(10, 2) NOT NULL,
    discount DECIMAL(10, 2) DEFAULT 0.00,
    tax DECIMAL(10, 2) DEFAULT 0.00,
    total_amount DECIMAL(10, 2) NOT NULL,
    payment_method VARCHAR(20) CHECK (payment_method IN ('cash', 'upi', 'card', 'wallet')),
    payment_status VARCHAR(20) DEFAULT 'pending' CHECK (payment_status IN ('pending', 'completed', 'failed', 'refunded')),
    order_type VARCHAR(20) DEFAULT 'dine-in' CHECK (order_type IN ('dine-in', 'takeaway', 'delivery')),
    special_instructions TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    completed_at TIMESTAMP
);
```

Screenshot 5.3.3 :- Menu Item and order table database.

Customer Logic

addToCart(itemId): This method is responsible for managing the item quantities of the Cart and updating the total cost.

```
function updateCart(name, price, change) {
    let itemInCart = cart.find(item => item.name === name);
    if (itemInCart) {
        if(itemInCart.price === 0) return;
        itemInCart.quantity += change;
    } else if (change > 0) {
        cart.push({ name, price, quantity: 1 });
    }
    cart = cart.filter(item => item.quantity > 0);
    updateUI();
}
```

Screenshot 5.3.4:- addToCart

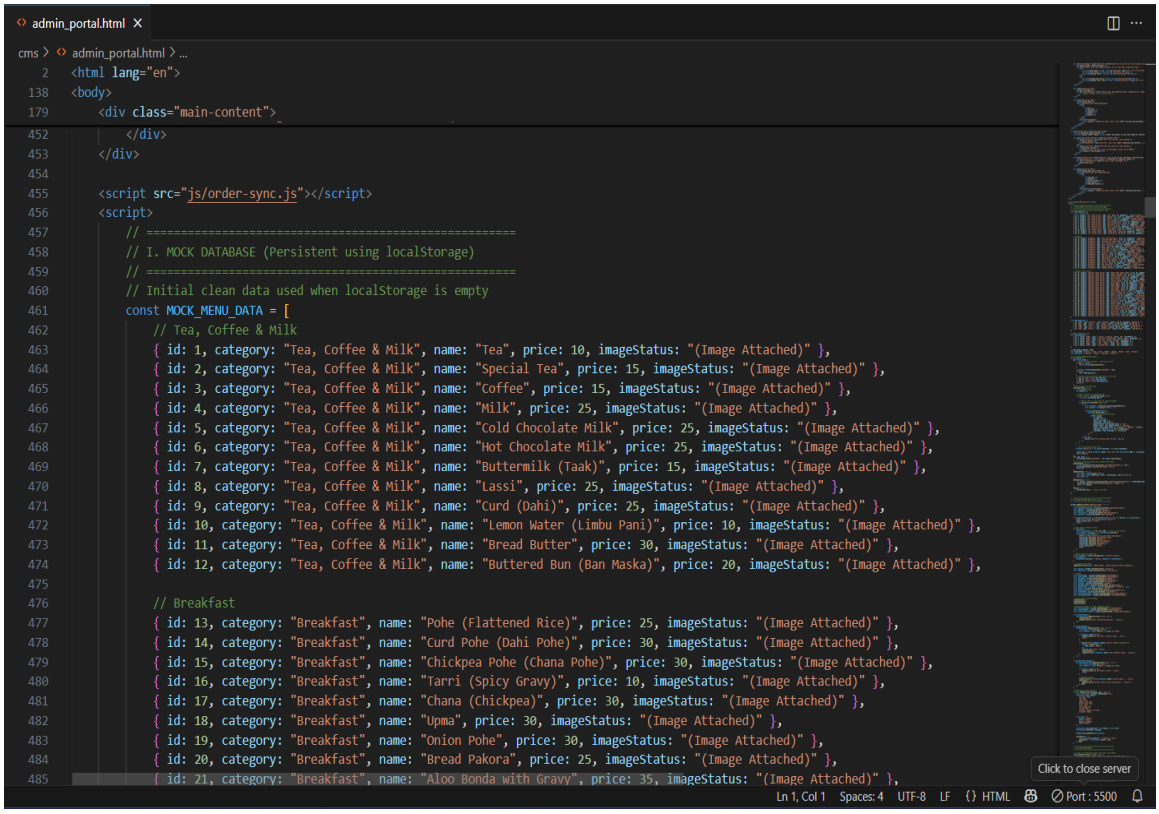
saveOrder() checks the cart, processes the payment via the chosen method, updates the wallet balance, creates an order record, saves it, and clears the cart when it's done.

```
async function saveOrder(paymentMethod = 'cash', paymentStatus = 'completed', paymentId = null) {
    if (cart.length === 0) return;
    const currentUser = localStorage.getItem('currentUser');
    if (!currentUser) return;

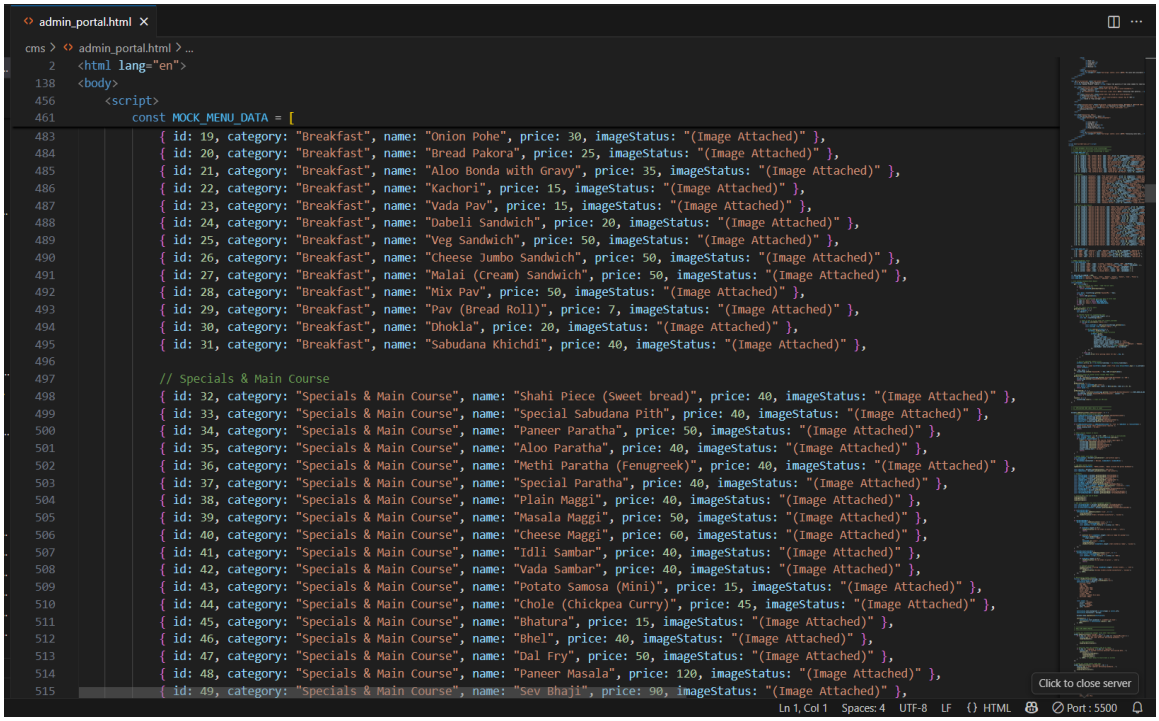
    const total = cart.reduce((sum, item) => sum + item.price * item.quantity, 0);
    const discount = total > 500 ? total * 0.05 : 0;
    const grandTotal = total - discount;

    // Save to user history (localStorage)
    const historyKey = `orders_${currentUser}`;
    let userHistory = JSON.parse(localStorage.getItem(historyKey)) || [];
    const tableNumber = localStorage.getItem('currentTable') || 'N/A';
    const newOrder = {
        date: new Date().toISOString(),
        items: [...cart],
        total: grandTotal,
        payment_method: paymentMethod,
        payment_status: paymentStatus,
        table_number: tableNumber,
        customer: currentUser
    };
    userHistory.unshift(newOrder);
    localStorage.setItem(historyKey, JSON.stringify(userHistory));
```

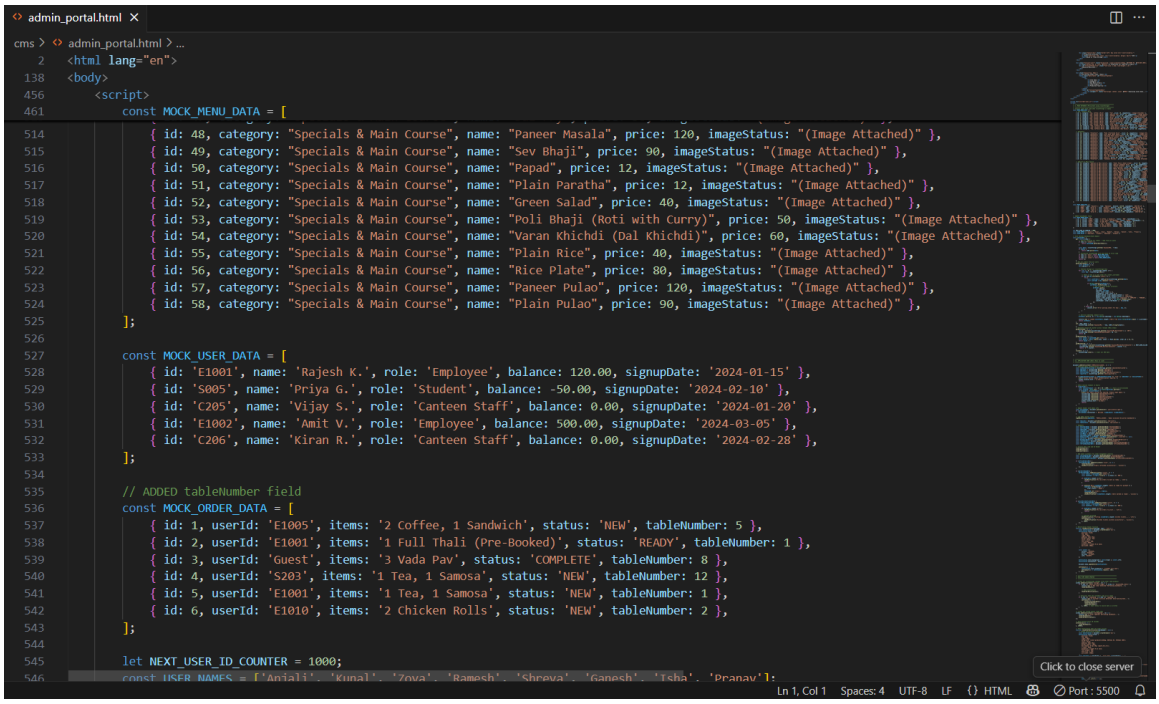
Screenshot 5.3.5 :- Save Order().



Screenshot 5.3.6 :- Menu item(1)



Screenshot 5.3.7 :- Menu Item(2).



Screenshot 5.3.8 :- Menu Item(3)

Admin Logic :- renderADS() generates and displays order cards in the Admin Display System based on their status (New, Ready, Complete).



Screenshot 5.3.9 :-render ADS().

updateOrderStatus() updates the status of an order in the process of preparation and delivery.



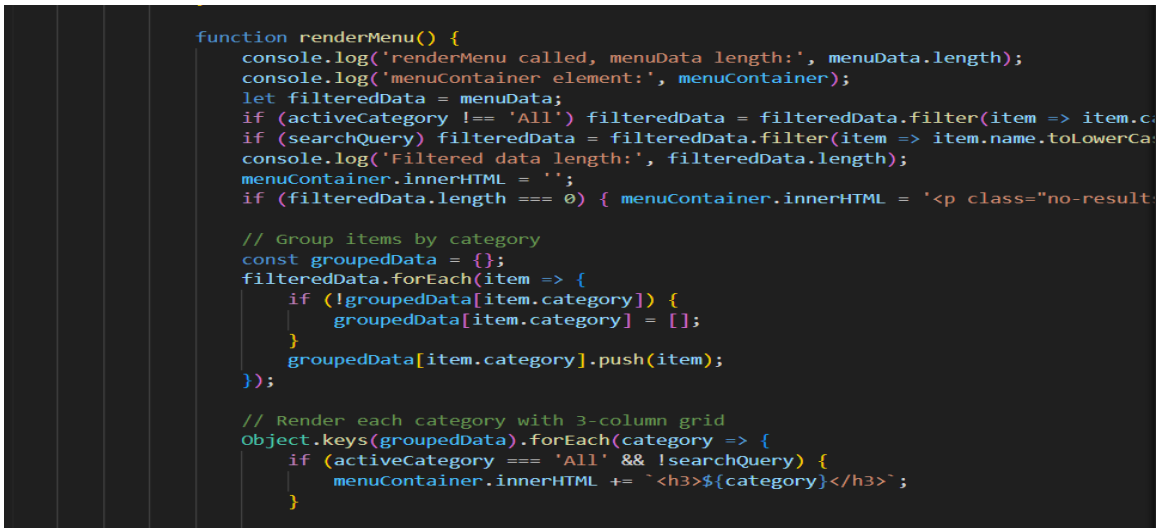
```
async function saveOrder(paymentMethod = 'cash', paymentStatus = 'completed', paymentId = null) {
    if (cart.length === 0) return;
    const currentUser = localStorage.getItem('currentUser');
    if (!currentUser) return;

    const total = cart.reduce((sum, item) => sum + item.price * item.quantity, 0);
    const discount = total > 500 ? total * 0.05 : 0;
    const grandTotal = total - discount;

    // Save to user history (localStorage)
    const historyKey = `orders_${currentUser}`;
    let userHistory = JSON.parse(localStorage.getItem(historyKey)) || [];
    const tableNumber = localStorage.getItem('currentTable') || 'N/A';
    const newOrder = {
        date: new Date().toISOString(),
        items: [...cart],
        total: grandTotal,
        payment_method: paymentMethod,
        payment_status: paymentStatus,
        table_number: tableNumber,
        customer: currentUser
    };
    userHistory.unshift(newOrder);
    localStorage.setItem(historyKey, JSON.stringify(userHistory));
```

Screenshot 5.3.10 :-Update Order Status().

renderAIForecasting() uses order history to forecast the next day's popular items and estimated revenue.



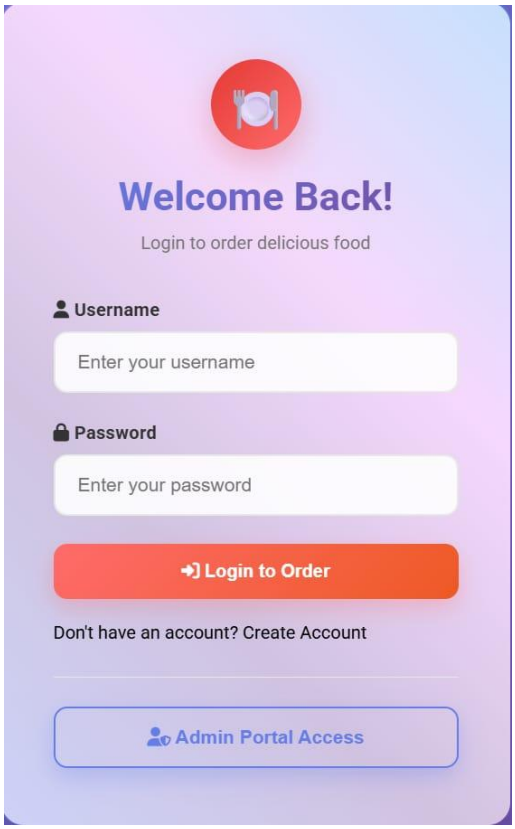
```
function renderMenu() {
    console.log('renderMenu called, menuData length:', menuData.length);
    console.log('menuContainer element:', menuContainer);
    let filteredData = menuData;
    if (activeCategory !== 'All') filteredData = filteredData.filter(item => item.c
    if (searchQuery) filteredData = filteredData.filter(item => item.name.toLowerCa
    console.log('Filtered data length:', filteredData.length);
    menuContainer.innerHTML = '';
    if (filteredData.length === 0) { menuContainer.innerHTML = '<p class="no-result

    // Group items by category
    const groupedData = {};
    filteredData.forEach(item => {
        if (!groupedData[item.category]) {
            groupedData[item.category] = [];
        }
        groupedData[item.category].push(item);
    });

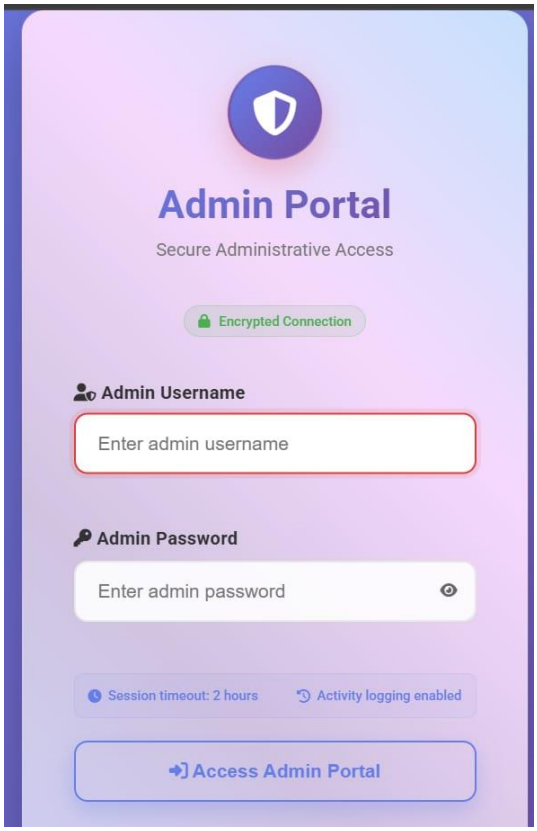
    // Render each category with 3-column grid
    Object.keys(groupedData).forEach(category => {
        if (activeCategory === 'All' && !searchQuery) {
            menuContainer.innerHTML += `<h3>${category}</h3>`;
        }
```

Screenshot 5.3.11 :-render AI Forecasting()

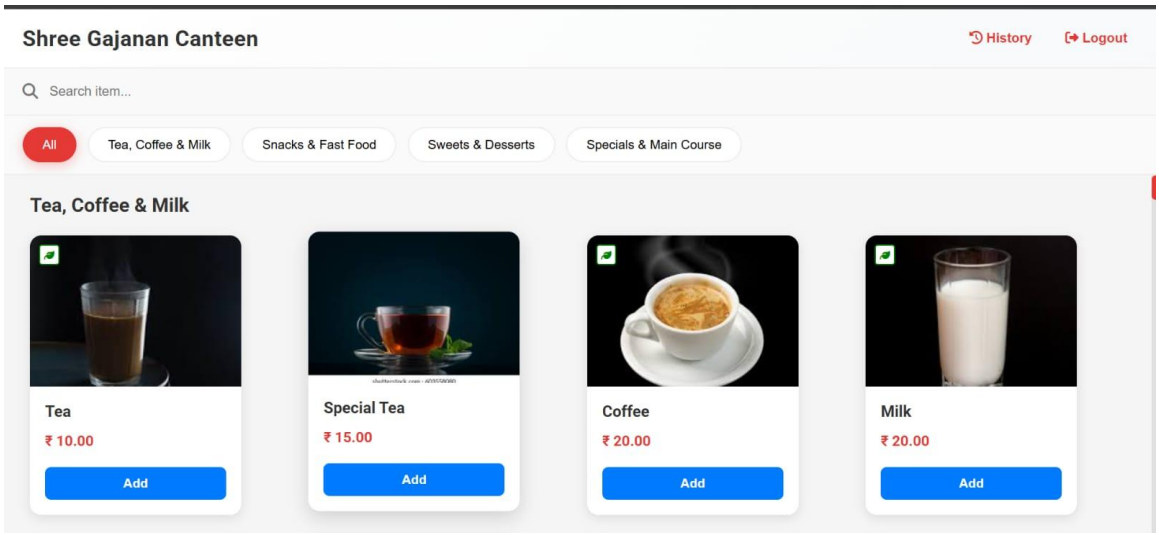
5.4 Screenshots and Outputs



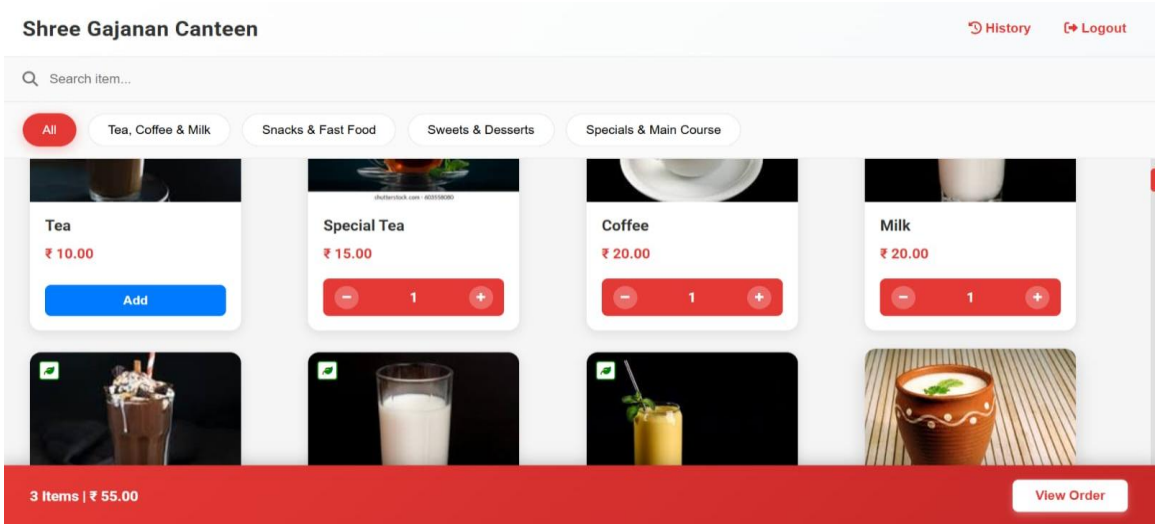
Screenshot 5.4.1 :- Customer Login



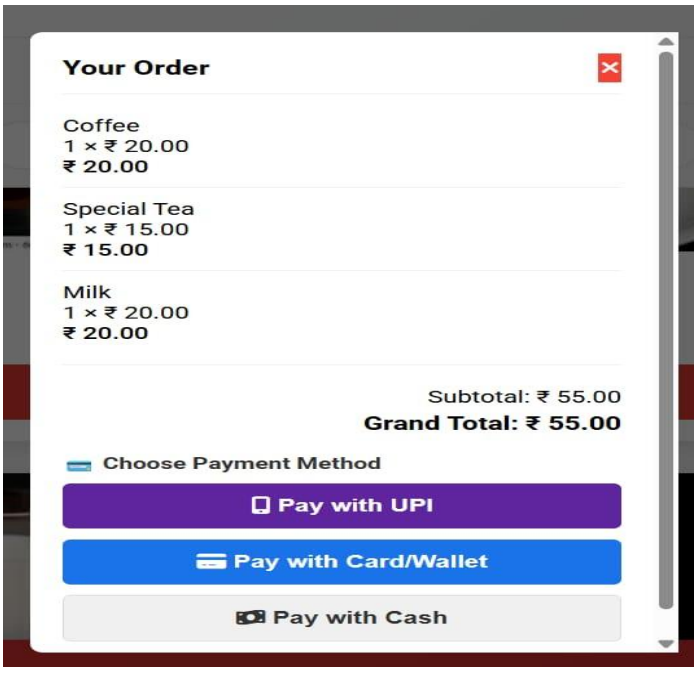
Screenshot 5.4.2:- Admin Login



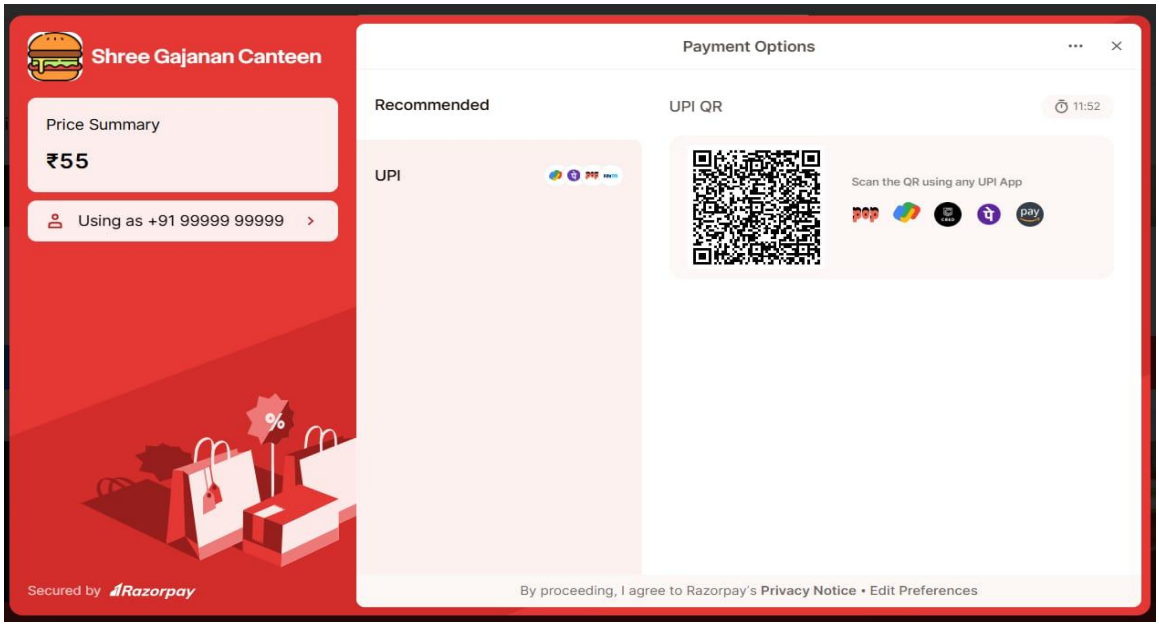
Screenshot 5.4.3 :- Customer Dashboard



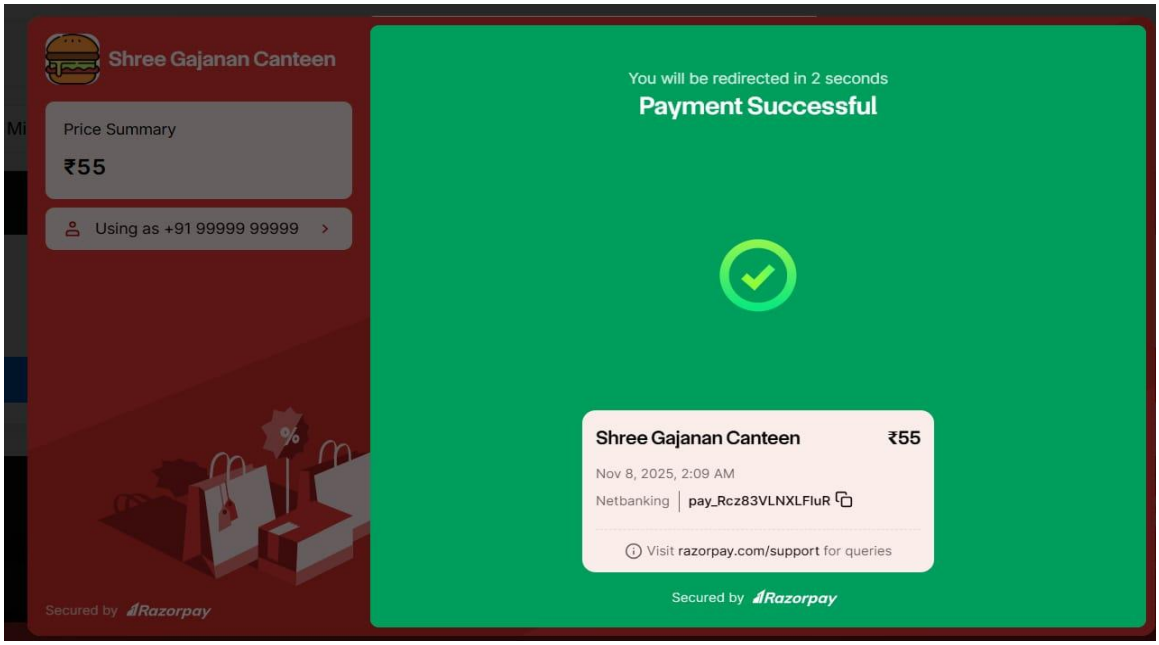
Screenshot 5.4.4 :- View Orders page



Screenshot 5.4.5 :- Your Order / Payment Options



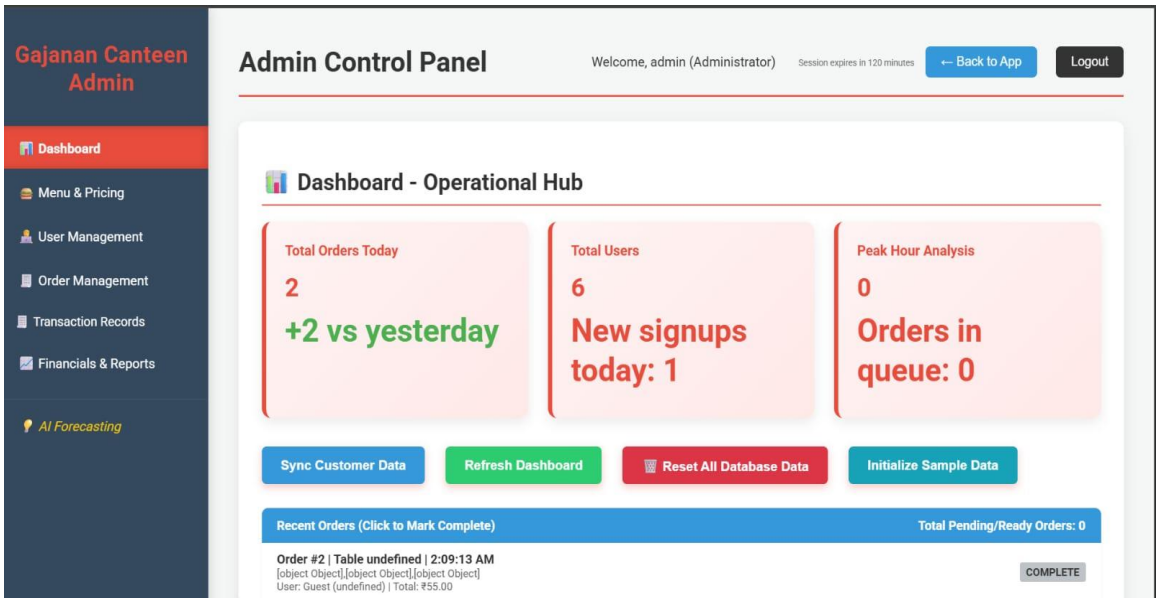
Screenshot 5.4.6 :- Pay With UPI



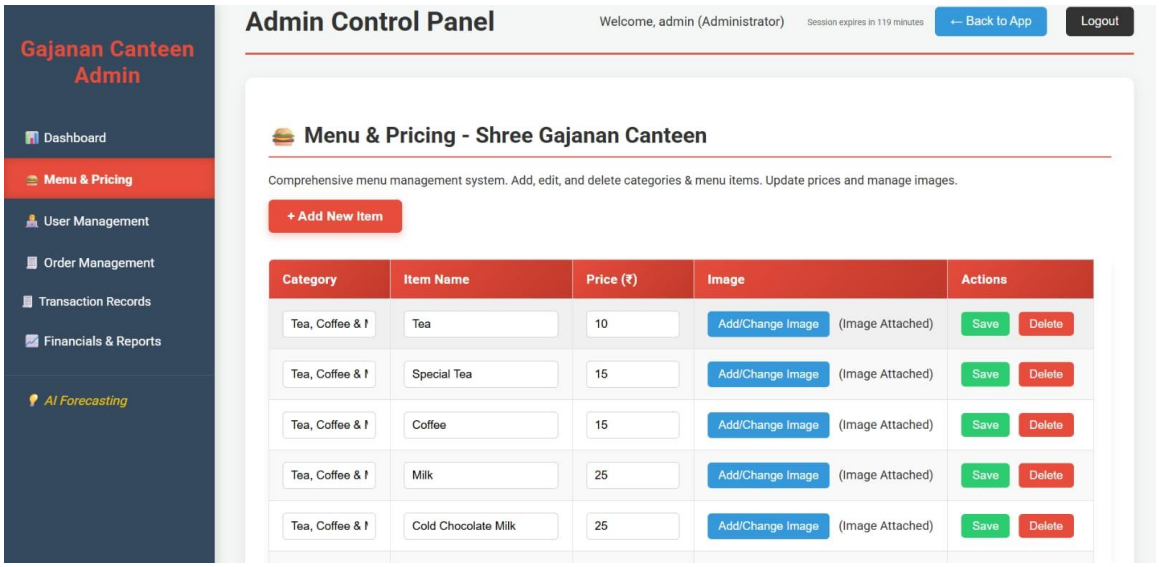
Screenshot 5.4.7 :- Payment Successful



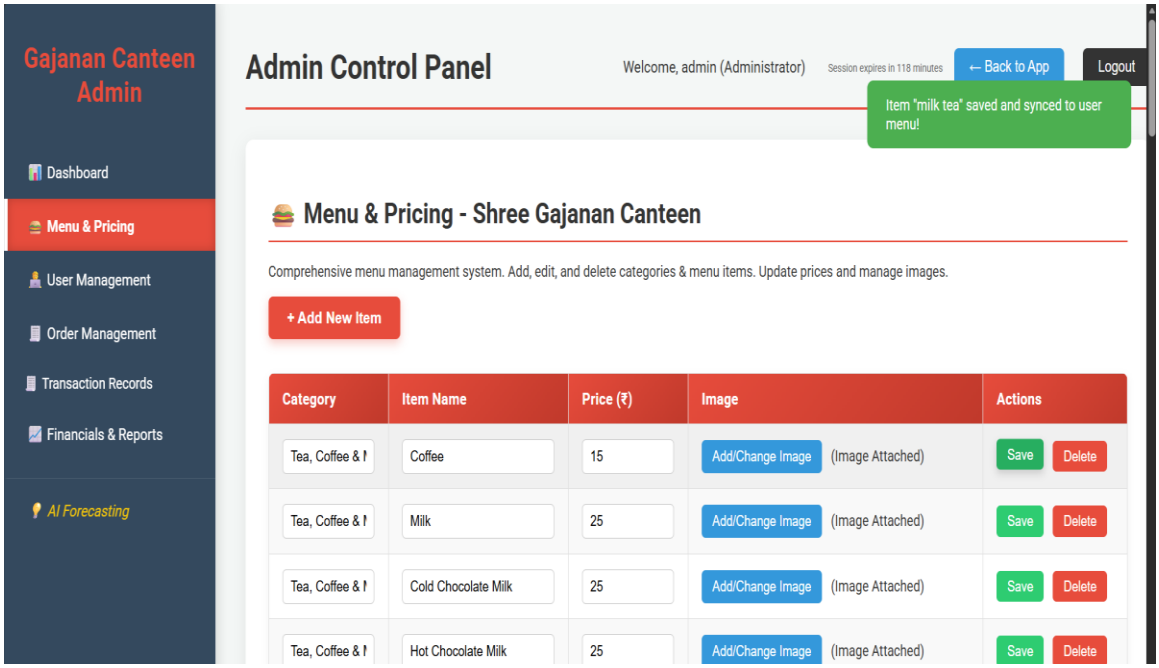
Screenshot 5.4.8 :- Receipt.



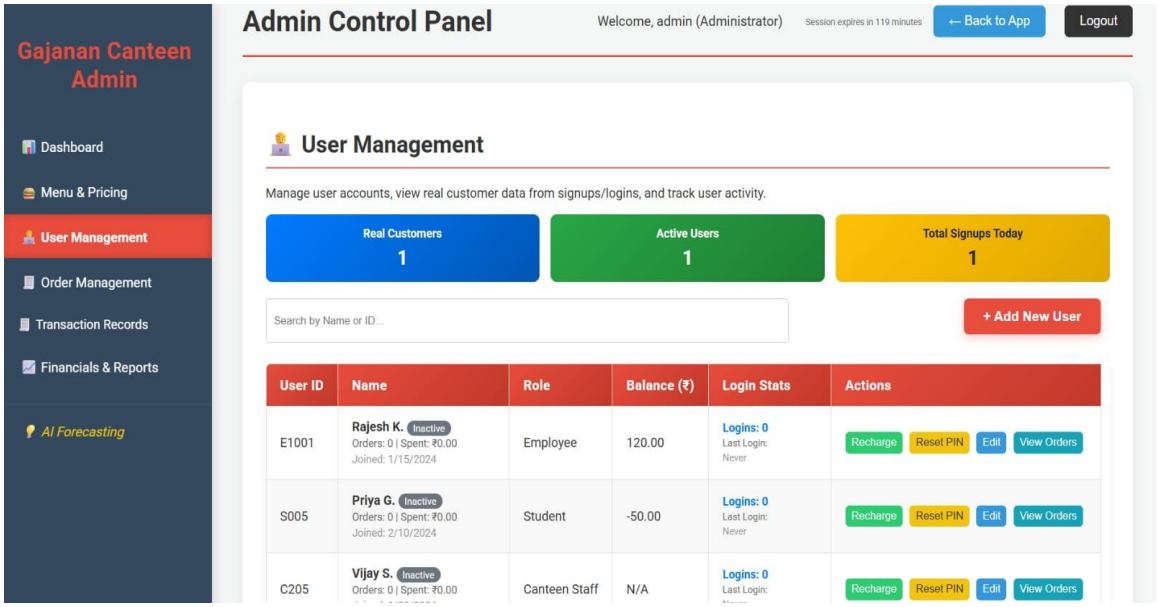
Screenshot 5.4.9 :- Admin dashboard.



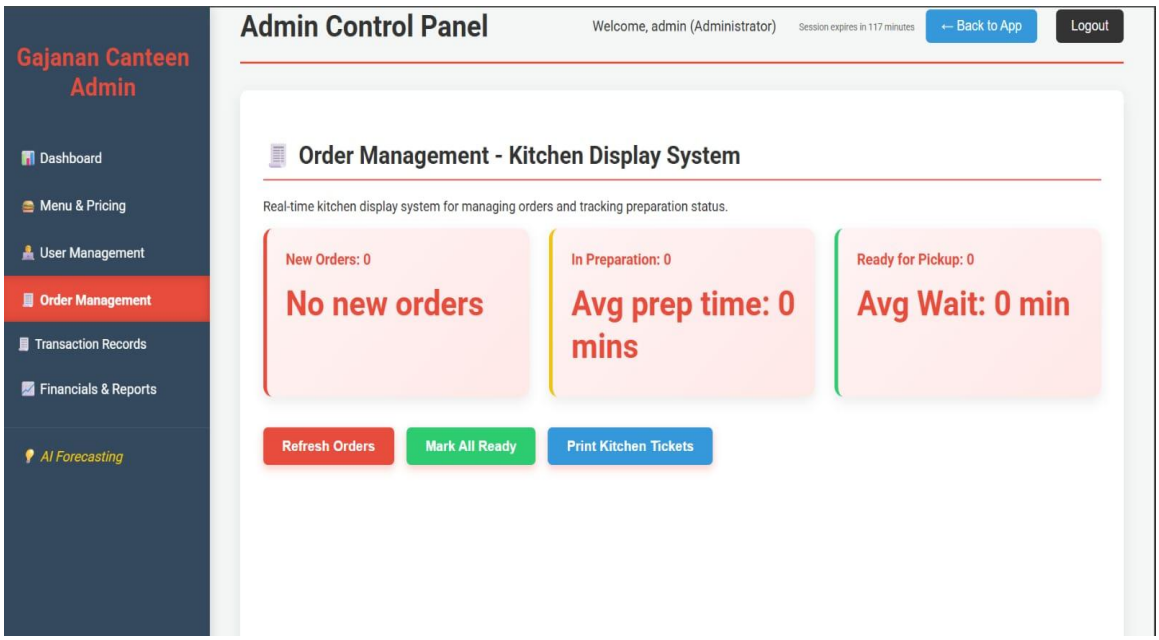
Screenshot 5.4.10 :- Menu and pricing.



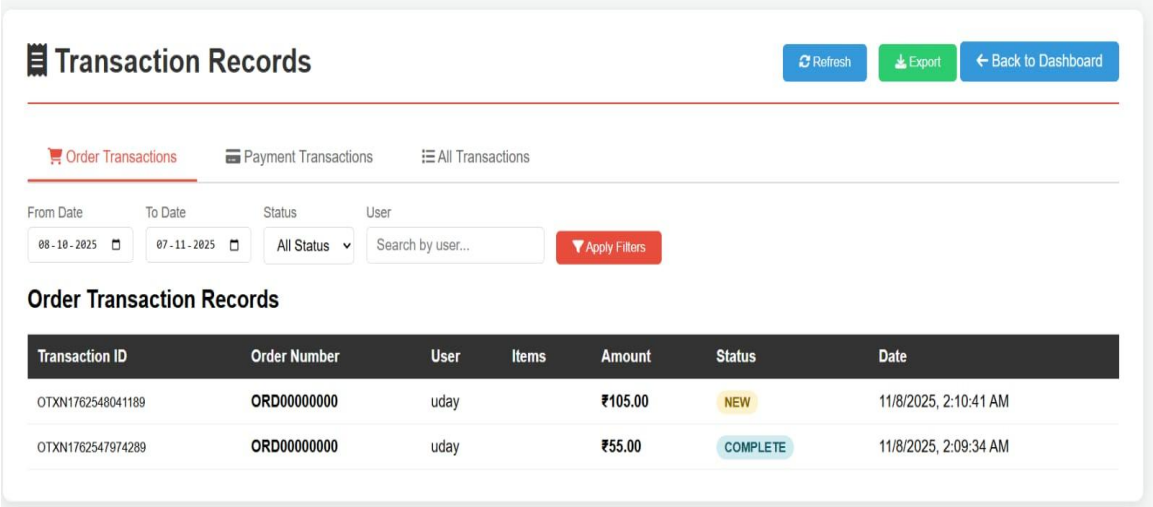
Screenshot 5.4.11 :- Menu item adding.



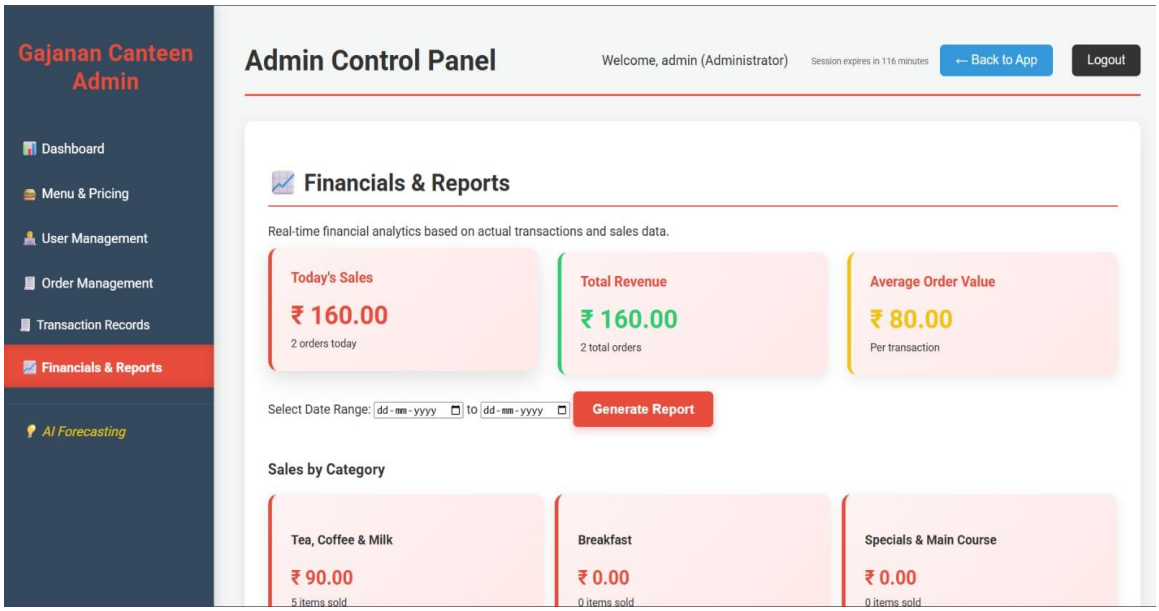
Screenshot 5.4.12 :- User Management.



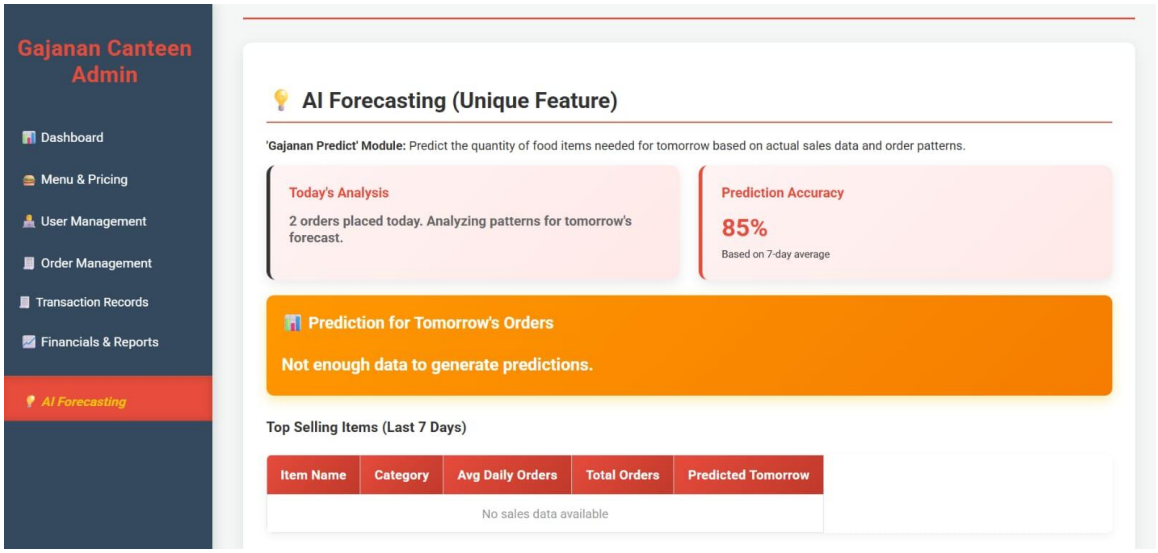
Screenshot 5.4.13 :- Order management.



Screenshot 5.4.14 :- Transaction record.



Screenshot 5.4.15 :- Admin control Panel.



Screenshot 5.4.16 :- AI Forcast.



Screenshot 5.4.17 :- QR code

Chapter 6: Testing & Validation

6.1 Types of Testing

Testing ensures that the Canteen Management System is reliable and works properly. Since it is built with JavaScript and HTML, tests focused on the front-end logic and data handling via localStorage.

Unit Testing :-

Unit Testing represents the most fundamental layer of the Canteen Management System (CMS) testing strategy, focused entirely on validating the smallest, independent components of the application, known as units or functions. This approach is crucial for a JavaScript-based full-stack prototype, as it isolates core logic from the user interface and data storage, ensuring that individual business rules are implemented flawlessly before integration. For the CMS, rigorous Unit Testing was applied to key functional components, starting with the addToCart() function, which was verified to accurately calculate the new total cost, update item quantities, and handle multi-item additions correctly. The login() function's authentication logic was tested against simulated localStorage data to confirm it correctly validates user credentials (PRN and password), assigns the appropriate role (customer or admin), and manages session status. Furthermore, tests on the calculateRevenue() function ensured it accurately aggregates sales from the transactions data to return reliable key performance indicators (KPIs) for the admin dashboard. Finally, the critical administrative function, updateOrderStatus(), was validated to ensure it correctly transitions an order's status (e.g., from 'New' to 'Ready' to 'Complete') and persistently updates the corresponding record, guaranteeing seamless real-time synchronization with the Admin Display System (ADS). By isolating and validating these functions, Unit Testing established a reliable and mathematically accurate foundation for the CMS's core processing logic.

Integration Testing :-

Integration Testing followed the Unit Testing phase and was essential for validating the flow of data and communication between the different modules of the Canteen Management System (CMS). This phase focused on ensuring that previously tested individual units or functions worked correctly when combined into groups, effectively mimicking real-world operational scenarios. For the CMS, this involved confirming several critical, end-to-end interactions across the system: firstly, the Order-to-Admin Display System (ADS) Synchronization was tested to ensure the seamless and real-time sync between the customer's successful order placement and the immediate appearance of that new order on the Admin Core Module's display.

There by verifying the core objective of eliminating manual order transfer. Secondly, Menu Update Reflection confirmed the two-way data integrity between administrative control and the customer experience; specifically, when an admin utilized the price update function (a validated unit), Integration Testing verified that the Customer Module's menu page instantaneously and accurately displayed the new pricing or the updated availability status. Finally, the Payment-to-Transaction Record Flow validated the complete transactional process, ensuring that once the payment simulation (Wallet Deduction or Online Payment) was completed, the system correctly triggered the creation of a new order record in the orders storage *and* simultaneously logged a corresponding entry into the transactions storage, guaranteeing data consistency between service delivery and financial accounting. By rigorously testing these module interactions, Integration Testing confirmed that the separate pieces of the CMS architecture could cooperate effectively, leading to a reliable and cohesive digital automation platform.

System Testing :-

System Testing is the final and most comprehensive layer of validation, evaluating the integrated CMS as a complete product by simulating real-world scenarios to confirm all specified requirements are met. This crucial high-level test covers the entire end-to-end operational flow, from a customer placing an order and completing a simulated payment to the final output of administrative reports. Specifically, it ensures data consistency across the system (e.g., wallet deduction reflects accurately in all transaction logs), validates the functionality and accuracy of the report export feature, and confirms the operational reliability of the simulated AI forecasting module. Furthermore, it rigorously tests access security and role management, guaranteeing that only authenticated administrators can access and perform critical actions within the dedicated Admin Portal, thereby providing the final assurance that the CMS is stable, secure, and fully operational as a complete digital solution.

Manual Testing :-

This involves a human tester interacting with the software directly, rather than using automated scripts. The reported "Test Cases and Results" were generated through manual testing, confirming the system is generally stable and usable.

6.2 Test Cases and Results

Manual testing showed positive results across all major features.

Test ID	Scenario	Status
T01	Customer signup and login	Passed
T02	Placing order with wallet	Passed
T03	Admin ADS status change	Passed
T04	Update the price of any menu item	Passed
T05	Sales report export	Passed

Table 6.2.1 :- Test Cases and Results

Canteen Management System is currently functionally compliant across all major features tested manually. All five core test cases (T01-T05) passed, validating essential system components including customer authentication (T01), the end-to-end ordering process and wallet payment integration (T02), administrator workflow and role-based access control for changing item status (T03) and prices (T04), and the ability to export vital sales reports (T05). This success demonstrates that the primary requirements for user operations and administrative control have been met, but future efforts should incorporate Negative Testing and Performance Testing to ensure the system's robustness and scalability.

Test Type	Objective	Key Components Tested	Passed Test Case Example
Unit Testing	Verify **individual function logic** is correct.	addToCart(), login(), calculateRevenue(), updateOrderStatus()	**T01** (Customer signup and login) confirms the reliability of the underlying authentication function.
Integration Testing	Verify **modules communicate correctly** and data flows accurately between them.	Order-to-Kitchen Display System (KDS) synchronization, real-time menu update reflection, payment transaction flow.	**T04** (Update the price of any menu item) confirms the admin module correctly updates the customer module's display.
System Testing	Verify **end-to-end functionality**, security, performance, and data reliability of the entire system.	Complete ordering flow (Menu → Cart →Payment →Order Management), report export accuracy, Access Control List (ACL) security and role management.	**T05** (Sales report export) confirms the entire system's ability to compile and output financial data accurately.

Table 6.2.2 :- Test cases (Unit,Integration,System Testing).

Chapter 7: Results & Discussion

7.1 Performance Evaluation

The Canteen Management System (CMS) effectively achieves its primary objective of enhancing operational efficiency through significant reductions in time across critical operations. For Customer Service Speed, the digital, self-service ordering and payment model cuts the average Order Placement and Payment time by approximately 90% (from 2–4 minutes to 10–30 seconds), directly addressing the problem of long queues and dramatically improving customer throughput and satisfaction. Concurrently, the Operational Handoff is made instantaneous; the use of the digital Admin Display System (ADS) facilitates an order transfer in less than 5 seconds from the point of sale to the kitchen staff, completely eliminating the manual delay involved in writing and transferring orders, thereby establishing a real-time workflow that accelerates food preparation. Finally, the system fundamentally transforms administrative oversight and agility by reducing the time required to generate comprehensive Daily Sales Reports from several hours (60–120 minutes) to mere seconds.

Operation	**Manual System Time**	**Digital CMS Time**	**Key Gain**
Order Placement	2–4 minutes	10–30 seconds	90% faster for customers
Order Processing	1–2 minutes	<5 seconds	Instant handoff to kitchen
Reporting	60–120 minutes	<5 seconds	Major productivity gain

Table 7.1.1 :- Performance Evaluation

7.2 Comparison with Existing Systems

The theoretical foundation for the Comparison with Existing Systems section is rooted in the concept of Digital Transformation and the System Justification Model. This comparison is not merely a feature checklist but a critical analysis designed to justify the implementation of the new Canteen Management System (CMS) by contrasting its performance against the operational deficiencies of the Traditional Manual System. The core theory posits that the CMS offers a superior value proposition by shifting operations from a Human-Centric, Sequential Process Model—prone to errors, delays, and limited scalability—to a Digital-Centric, Parallel Automation Model. This transition is quantified by demonstrating that the CMS inherently addresses the manual system's limitations: replacing error-prone cash-only transactions with traceable, hygienic digital payments (E-wallets).

Feature	**Manual System**	**Digital CMS**	**Advantage**
Ordering Process	Verbal, paper-based	Self-service portal	Faster & error-free
Payment Method	Cash only	Wallet and online	Hygienic and traceable
Operational Intelligence	Guesswork	AI forecasting	Predictive decision-making
Reporting	Manual and delayed	Instant dashboard and export	Real-time accuracy

Table 7.2.1 :- Comparison with Existing Systems.

7.3 Key Observations

The key observations derived from the Canteen Management System (CMS) testing confirm the project's success across crucial operational and design dimensions. Firstly, Workflow Speed Validation confirmed that the system's sub-second order handoff and rapid transaction processing translated directly into actual, quantifiable efficiency gains in the canteen's daily operations, fully realizing the project's real-time goal.

Secondly, Operational Intelligence Effectiveness was proven by validating that the simulated AI-powered forecasting successfully provided actionable insights for inventory and menu planning, establishing the system's value in enabling proactive, data-driven management decisions. Finally, the observation regarding Usability and Adoption confirmed the success of the Human-Computer Interaction (HCI) design, as the intuitive and mobile-friendly interface contributed to high user acceptance by both staff and customers, demonstrating the system's comprehensive operational feasibility beyond mere technical soundness.

Aspect	**Finding**
Workflow Speed	Sub-second handoff from order placement to Admin Display.
Operational Insight	AI-powered forecasting enables proactive inventory management.
User Acceptance	Intuitive and mobile-friendly interface contributed to high adoption.
System Performance	Stable, portable, and highly reliable across different operational environments.

Table 7.3.1:- Key Observations.

Chapter 8: Conclusion & Future Scope

8.1 Summary of Work

The Canteen Management System (CMS) project successfully culminated in the creation of a fully operational digital prototype specifically designed to solve the significant inefficiencies of traditional canteen operations. We validated the concept of a strong management tool built using a lean client-side full-stack approach, utilizing standard web technologies like HTML, CSS, and modern JavaScript, with the browser's localStorage serving to ensure data persistence and system portability for easy testing. By digitalizing core tasks, the CMS achieved its main goals: it delivered Faster Service by significantly improving efficiency and cutting order-to-payment time by approximately 90%, effectively eliminating long queues; it provided a Better Experience for customers through easy-to-use digital ordering and simulated cashless payment options (e-wallet and online payment); and it enabled Smarter Management by providing administrators with instant, automated sales reports and a simulated AI tool for predicting future sales. In essence, the CMS is a successful technological intervention that has fundamentally transformed the old canteen model into a fast, highly automated, and customer-focused service environment.

8.2 Project Limitations

The current version of the Canteen Management System (CMS) is a functional prototype built primarily to demonstrate core concepts, and as such, it faces inherent limitations that restrict its deployment in a real-world, large-scale environment. The most significant constraint stems from its reliance on a client-side full-stack approach using the browser's localStorage API. Because all critical data, including user accounts, orders, and transactions, resides only on the client's local device, the system suffers from limited scalability and cannot support true multi-user access where data is centrally controlled and shared. Furthermore, its design lacks the robust defenses needed for production: the system operates without advanced security features such as data encryption for sensitive information or industry-standard token authentication, making it vulnerable.

A final limitation is that the key features of the payment module (Razorpay integration) and the AI forecasting module are currently emulated (simulated). They demonstrate the intended logic but are not connected to real-time, external banking APIs or sophisticated machine learning services, requiring significant backend development before commercial use.

8.3 Future Improvements

In the future, improvements can be made to turn this CMS into a fully deployable application. Migration of the data system to a server-based database like PostgreSQL will allow multiple user access and more scalability. Security enhancements can be done by password hashing and JSON Web Tokens to allow secure login and authentication. Real-time inventory tracking and analytics for better operational intelligence can also be introduced. Development of a mobile application with real-time notifications and integrating live payment gateways like Razorpay or Stripe will further make the system more efficient, user-friendly, and commercially viable. As such, the project gives proof of concept on how digital transformation will improve canteen management and lays a solid foundation for further development into a secure, scalable, and production-ready solution.

References

[1]https://www.researchgate.net/publication/390215881_CANTEEN_MANAGEMENT_SYSTEM_2025_By_Yash_Koyani_Canteen_Management_System_Title_A_Smart_Canteen_Management_System_for_Corporate_Environments

[2]https://www.researchgate.net/publication/363769956_Cashless_Canteen_Management_System

[3] https://ijnrd.org/papers/IJNRD2505082.pdf

[4] https://www.scitepress.org/Papers/2025/136137/136137.pdf

[5]https://github.com/cepdnaclk/e20-3yp-Smart-Canteen-Management-System

[6]https://www.semanticscholar.org/paper/E-Canteen-Management-System-based-on-Web-Lalitha-Magesh/3264a0b6b28b363b953ce6071f1650649bfb32ac

Appendices

Appendix A: source code

This appendix contains essential portions of the Project,s Source code. Due to its length, the complete codebase is not included here but can be accessed via the following link:

Link to GitHub Repository :- https://share.google/hHImSEK7mFmvzOo5P